

Largest Concatenation-Component in Regular Expressions

Hyperscan implements algorithms to decompose regexes into literals and FA in its Rose engine. There may be a relation between how well a regex can be decomposed and the size of its largest concatenation-component.

1.1 Recurrences

Let $\mathcal{A}$ be a finite alphabet and $\mathcal{T}$ be the set of all regexes excluding ϵ over $\mathcal{A}$. The size function $|\cdot|$ is the number of nodes.

Definition 1 (BST-like distribution for regexes). *We define $p : \mathcal{T} \rightarrow [0, 1]$ as follows.*

1. $\sum_{c \in \mathcal{A}} p(c) = 1,$
2. $p \left(\begin{array}{c} \bullet \\ \swarrow \searrow \\ T_1 \quad T_2 \end{array} \right) = \frac{u}{n-2} p(T_1) p(T_2), \text{ where } |T_1| + |T_2| = n - 1.$
3. $p \left(\begin{array}{c} + \\ \swarrow \searrow \\ T_1 \quad T_2 \end{array} \right) = \frac{v}{n-2} p(T_1) p(T_2), \text{ where } |T_1| + |T_2| = n - 1.$
4. $p \left(\begin{array}{c} * \\ | \\ T \end{array} \right) = (1 - u - v) p(T).$

We want to compute the expected size of the largest all-concatenation component in a regex over all of equivalent forms of its parse tree. The reason is as follows. Take the regex

$$(abc + d)(xy + z) = abcxy + abcxz + dxy + dz.$$

We expect the largest concatenation-component to be in $abcxy$ instead of abc because the former is extracted by Hyperscan. Therefore, a subtree may *yield* its concatenation-component to its parent if the parent is a concatenation node.

Recall the construction

$$\mathcal{T} = \sum_{c \in \mathcal{A}} c + \begin{array}{c} * \\ | \\ \mathcal{T} \end{array} + \begin{array}{c} \bullet \\ \swarrow \searrow \\ \mathcal{T} \quad \mathcal{T} \end{array} + \begin{array}{c} + \\ \swarrow \searrow \\ \mathcal{T} \quad \mathcal{T} \end{array}.$$

Define the random variables

1. M_n : the largest concatenation-component that is not inside a Kleene star of a regex of size n *itself*,
2. Y_n : the largest concatenation-component that is not inside a Kleene star of a regex of size n that it *yields* to its parent.

We have the following recurrences

$$Y_{n+2} = \begin{cases} Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 & \text{prob. } u, \\ \max(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)}) & \text{prob. } v, \\ 0 & \text{prob. } 1 - u - v \end{cases} \quad Y_1 = Y_2 = 0. \quad (1.1)$$

$$M_{n+2} = \begin{cases} \max(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1, M_{U_n}^{(1)}, M_{n+1-U_n}^{(2)}) & \text{prob. } u, \\ \max(M_{U_n}^{(1)}, M_{n+1-U_n}^{(2)}) & \text{prob. } v \\ 0 & \text{prob. } 1 - u - v \end{cases} \quad M_1 = M_2 = 0. \quad (1.2)$$

We have used $Y_n^{(1)}, Y_n^{(2)}, M_n^{(1)}, M_n^{(2)}$ as independent copies of Y_n and M_n respectively, and U_n as a uniform random variable over $\{1, 2, \dots, n\}$ independent of all $Y_n^{(i)}$ and $M_n^{(i)}$.

1.2 The yield recurrence

We rewrite the recurrence (1.1) as

$$Y_{n+2} \stackrel{d}{=} I_n \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 \right) + J_n \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right). \quad (1.3)$$

In (1.3), (I_n, J_n) is a pair of mutually exclusive Bernoulli random variables with $\mathbb{E}[I_n] = u$ and $\mathbb{E}[J_n] = v$ such that $I_n + J_n \leq 1$. The sequences $\mathbf{Y}^{(1)} = (Y_n^{(1)})_{n \geq 1}$ and $\mathbf{Y}^{(2)} = (Y_n^{(2)})_{n \geq 1}$ are independent copies of the sequence $\mathbf{Y} = (Y_n)_{n \geq 1}$. The notion of equality is in distribution. Note that we can reuse the same variables $Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)}$ and U_n across both terms because at most one indicator is non-zero for any given realization. Let $\mathbf{I} = (I_n)_{n \geq 1}$, and define $\mathbf{J}$ and $\mathbf{U}$ similarly. We assume that the sequence of pairs $(\mathbf{I}, \mathbf{J})$ and the sequence $\mathbf{U}$ each consist of independent and identically distributed elements. Furthermore, the entire stochastic processes $(\mathbf{I}, \mathbf{J}), \mathbf{U}, \mathbf{Y}^{(1)}$, and $\mathbf{Y}^{(2)}$ are mutually independent.

1.2.1 Lower and Upper Bounds of the Expectation

Let $y_n = \mathbb{E}[Y_n]$. Take expectation on both sides, we have

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^n y_i + \frac{v}{n} \sum_{i=1}^n \mathbb{E}[\max(Y_i, Y_{n+1-i})].$$

For an upper bound, we use the fact that $\max(Y_i, Y_{n+1-i}) \leq Y_i + Y_{n+1-i}$, which gives an upper bound sequence

$$y_{n+2}^U = u + \frac{2(u+v)}{n} \sum_{i=1}^n y_i^U, \quad y_1^U = y_2^U = 0. \quad (1.4)$$

For a lower bound, we use

$$\begin{aligned} \mathbb{E}[\max(Y_i, Y_{n+1-i})] &= \mathbb{E} \left[\frac{1}{2} (Y_i + Y_{n+1-i} + |Y_i - Y_{n+1-i}|) \right] \\ &\geq \frac{1}{2} (\mathbb{E}[Y_i] + \mathbb{E}[Y_{n+1-i}]) + \frac{1}{2} |\mathbb{E}[Y_i] - \mathbb{E}[Y_{n+1-i}]| \\ &= \frac{1}{2} (y_i + y_{n+1-i} + |y_i - y_{n+1-i}|). \end{aligned}$$

This gives the lower bound sequence

$$y_{n+2}^L = u + \frac{2u+v}{n} \sum_{i=1}^n y_i^L + \frac{v}{2n} \sum_{i=1}^n |y_i^L - y_{n+1-i}^L|, \quad y_1^L = y_2^L = 0. \quad (1.5)$$

Remark. Another lower bound is obtained as follows. By Jensen's inequality

$$\mathbb{E}[\max(Y_i, Y_{n+1-i})] \geq \max(\mathbb{E}[Y_i], \mathbb{E}[Y_{n+1-i}]) = \max(y_i, y_{n+1-i}) = y_{\max(i, n+1-i)},$$

where the last equality holds because y_n is non-decreasing in n (*we have to prove it*). This gives the following lower bound sequence

$$y_{n+2}^L = u + \frac{2u}{n} \sum_{i=1}^n y_i^L + \frac{2v}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^n y_i^L, \quad y_1^L = y_2^L = 0. \quad (1.6)$$

Looking at the bounds (1.5), we know that when $2u + v > 1$, (y_n) grows at least polynomially.

1.2.2 Asymptotic of the Expectation

Consider the case $2u + v > 1$. By the bounds, there exists

$$\alpha = \inf \left\{ \beta : \frac{Y_n}{n^\beta} \xrightarrow{d} 0 \right\}.$$

We know that $\alpha \in (2u + v - 1, 2(u + v) - 1]$. We want to see if indeed $y_n \sim Cn^\alpha$ for some $C > 0$.

We try the contraction method [2, 3, 1]. Let us mention some conventions. For a random variable X , denote by $\mathcal{L}(X)$ its distribution. Let $(\mathcal{M}, d)$ be a metric space where $\mathcal{M}$ is a set of probability distributions of our random variables and the metric d is such that $d(\mathcal{L}(X), \mathcal{L}(Y)) = 0$ implies $X \stackrel{d}{=} Y$. The idea is as follows

1. Guess a normalization X_n of Y_n that converges in distribution.
2. Plug X_n into the recurrence. Assume that X_n converges, show that the limit equation has a unique solution X^* . This is done by proving that the operator T defined by the right-hand side of the limit equation is a contraction.
3. Prove that indeed $d(\mathcal{L}(X_n), \mathcal{L}(X^*)) \rightarrow 0$.

The method is use to obtain more information about the limit distribution of X_n rather than just its expectation. For example, the following is the recurrence for the number of comparisons in Quicksort where the first element is chosen as the pivot [1]

$$Y_n = Y_{U_n}^{(1)} + Y_{n-1-U_n}^{(2)} + n - 1.$$

One can find the asymptotic of $\mathbb{E}[Y_n]$ by taking expectation on both sides. The contraction is then used to show that $X_n = \frac{Y_n - \mathbb{E}[Y_n]}{n} \rightarrow X^*$, where the denominator n is the “guessed” variance.

Our case is weaker. The normalization is $X_n = Y_n/n^\alpha$ such that $X_n \xrightarrow{d} X^*$. Plugging into the recurrence (1.3), we have

$$\begin{aligned} (n+2)^\alpha X_{n+2} &\stackrel{d}{=} I \left(U_n^\alpha X_{U_n}^{(1)} + (n+1-U_n)^\alpha X_{n+1-U_n}^{(2)} + 1 \right) \\ &\quad + J \max \left(U_n^\alpha X_{U_n}^{(1)}, (n+1-U_n)^\alpha X_{n+1-U_n}^{(2)} \right) \end{aligned}$$

Equivalently,

$$\begin{aligned} X_{n+2} &\stackrel{d}{=} I \left(\left(\frac{U_n}{n+2} \right)^\alpha X_{U_n}^{(1)} + \left(\frac{n+1-U_n}{n+2} \right)^\alpha X_{n+1-U_n}^{(2)} + \frac{1}{(n+2)^\alpha} \right) \\ &\quad + J \max \left(\left(\frac{U_n}{n+2} \right)^\alpha X_{U_n}^{(1)}, \left(\frac{n+1-U_n}{n+2} \right)^\alpha X_{n+1-U_n}^{(2)} \right) \end{aligned} \quad (1.7)$$

Proposition 1. *The right-hand side of the above recurrence converges in distribution to*

$$I \left(U^\alpha X^{(1)} + (1 - U)^\alpha X^{(2)} \right) + J \max \left(U^\alpha X^{(1)}, (1 - U)^\alpha X^{(2)} \right) \quad (1.8)$$

Proof. We first show that

$$\left(\frac{U_n}{n+2} \right)^\alpha X_{U_n}^{(1)} + \left(\frac{n+1-U_n}{n+2} \right)^\alpha X_{n+1-U_n}^{(2)} \xrightarrow{d} U^\alpha X^{(1)} + (1 - U)^\alpha X^{(2)}.$$

By Lemma ?? and Slutsky's theorem,

$$\frac{U_n}{n+2} = \frac{U_n}{n} \cdot \frac{n}{n+2} \xrightarrow{d} U.$$

To apply the Continuous Mapping Theorem, we need to show that

$$\left(\frac{U_n}{n+2}, X_{U_n}^{(1)}, X_{n+1-U_n}^{(2)} \right) \xrightarrow{d} \left(U, X^{(1)}, X^{(2)} \right).$$

Since $U, X^{(1)}, X^{(2)}$ are independent, this is equivalent to showing that $\forall u, x_1, x_2 \in \mathbb{R}$

$$\mathbb{P} \left(U_n \leq \lfloor nu \rfloor, X_{U_n}^{(1)} \leq x_1, X_{n+1-U_n}^{(2)} \leq x_2 \right) \longrightarrow \mathbb{P}(U \leq u) \mathbb{P}(X^{(1)} \leq x_1) \mathbb{P}(X^{(2)} \leq x_2).$$

Indeed,

$$\begin{aligned} & \mathbb{P} \left(U_n \leq \lfloor nu \rfloor, X_{U_n}^{(1)} \leq x_1, X_{n+1-U_n}^{(2)} \leq x_2 \right) \\ &= \sum_{k=1}^{\lfloor nu \rfloor} \mathbb{P}(U_n = k) \mathbb{P} \left(X_k^{(1)} \leq x_1, X_{n+1-k}^{(2)} \leq x_2 | U_n = k \right) \\ &= \frac{1}{n} \sum_{k=1}^{\lfloor nu \rfloor} \mathbb{P} \left(X_k^{(1)} \leq x_1, X_{n+1-k}^{(2)} \leq x_2 \right) \quad (U_n \perp (X_k^{(1)}, X_{n+1-k}^{(2)})) \\ &= \frac{1}{n} \sum_{k=1}^{\lfloor nu \rfloor} \mathbb{P} \left(X_k^{(1)} \leq x_1 \right) \mathbb{P} \left(X_{n+1-k}^{(2)} \leq x_2 \right) \quad (X_k^{(1)} \perp X_{n+1-k}^{(2)}) \end{aligned}$$

Let F_n be the common CDF of $X_n^{(1)}$ and $X_n^{(2)}$, and F be the CDF of $X^{(1)}$. Let $m = \lfloor nu \rfloor$. The problem becomes showing that

$$\frac{1}{n} \sum_{k=1}^m F_k(x_1) F_{n+1-k}(x_2) \longrightarrow u F(x_1) F(x_2).$$

Equivalently,

$$\frac{m}{n} \left(\frac{1}{m} \sum_{k=1}^m (F_k(x_1) F_{n+1-k}(x_2) - F(x_1) F(x_2)) \right) \longrightarrow 0.$$

We have

$$\begin{aligned}
& \frac{1}{m} \sum_{k=1}^m |F_k(x_1)F_{n+1-k}(x_2) - F(x_1)F(x_2)| \\
& \leq \frac{1}{m} \sum_{k=1}^m |F_k(x_1) (F_{n+1-k}(x_2) - F(x_2))| + \frac{1}{m} \sum_{k=1}^m |F(x_2) (F_k(x_1) - F(x_1))| \\
& \leq \frac{1}{m} \sum_{k=1}^m |F_{n+1-k}(x_2) - F(x_2)| + \frac{1}{m} \sum_{k=1}^m |F_k(x_1) - F(x_1)| \quad (|F_n| \leq 1)
\end{aligned}$$

The second term goes to zero by the fact that $F_n(x_1) \rightarrow F(x_1)$ and Cesàro Means. We can write the first term as

$$\frac{1}{m} \sum_{k=n-m+1}^n |F_k(x_2) - F(x_2)|.$$

If $u = 1$, its convergence is similar to the second term. If $u < 1$, then $n - m + 1 \rightarrow \infty$. By pointwise convergence, $\forall \epsilon > 0, \exists N, \forall n \geq N : |F_n(x_2) - F(x_2)| < \epsilon$. Hence, for $n - m + 1 \geq N$,

$$\frac{1}{m} \sum_{k=n-m+1}^n |F_k(x_2) - F(x_2)| < \epsilon.$$

Using Slutsky's theorem again, we have

$$\left(\frac{n+1}{n+2}, \frac{U_n}{n+2}, X_{U_n}^{(1)}, X_{n+1-U_n}^{(2)} \right) \xrightarrow{d} \left(1, U, X^{(1)}, X^{(2)} \right).$$

Now define the function $f : \mathbb{R}_+^4 \cap \{(t, u, x, y) : t \geq u\} \rightarrow \mathbb{R}$ by

$$f(t, u, x, y) = u^\alpha x + (t - u)^\alpha y.$$

Then f is continuous. The Continuous Mapping Theorem applies. The convergence

$$\max \left(\left(\frac{U_n}{n+2} \right)^\alpha X_{U_n}^{(1)}, \left(\frac{n+1-U_n}{n+2} \right)^\alpha X_{n+1-U_n}^{(2)} \right) \xrightarrow{d} \max \left(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right)$$

is similar. And finally, we define the continuous function $g(i, j, u, v) = iu + jv$ on $\mathbb{R}^4$ to finish the proof. □

Proposition 2. *Let $T : \mathcal{M} \rightarrow \mathcal{M}$ be an operator in the space of probability distributions with finite expectation, defined by*

$$T(\mu) = \mathcal{L} \left(I(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}) + J \max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) \right),$$

where $X^{(1)}, X^{(2)} \stackrel{iid}{\sim} \mu$, U is uniformly distributed on $[0, 1]$, I and J are Bernoulli with expectation u and v respectively, such that $I + J \leq 1$. The variables U, I, J and $(X^{(1)}, X^{(2)})$ are mutually independent. Then there exists $p \geq 2$ such that T is a contraction in the $(\mathcal{M}_p, \ell_p)$, where the ℓ_p metric is defined by

$$\ell_p(\mu, \nu) = \inf \{ \mathbb{E}[|X - Y|^p] : X \sim \mu, Y \sim \nu \}.$$

Proof. Let $\mu, \nu \in \mathcal{M}_p$. Let $X, X^{(1)}, X^{(2)} \stackrel{\text{iid}}{\sim} \mu$ and $Y, Y^{(1)}, Y^{(2)} \stackrel{\text{iid}}{\sim} \nu$ be independent of each other. We have

$$\begin{aligned} \ell_p(T(\mu), T(\nu)) &= \mathbb{E} \left[\left| I(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}) + J \max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) \right. \right. \\ &\quad \left. \left. - I(U^\alpha Y^{(1)} + (1-U)^\alpha Y^{(2)}) - J \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)}) \right| \right] \\ &\leq \mathbb{E} \left[\left| I \left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)} \right) \right| \right] \\ &\quad + \mathbb{E} \left[\left| J \left(\max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) - \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)}) \right) \right| \right] \end{aligned}$$

Consider the first term. We have

$$\begin{aligned} &\mathbb{E} \left[\left| I \left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)} \right) \right| \right] \\ &= \mathbb{E} \left[I \left| U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)} \right| \right] \quad (I \geq 0) \\ &= u \mathbb{E} \left[\left| U^\alpha (X^{(1)} - Y^{(1)}) + (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right] \quad (\text{independence of } I) \\ &\leq u \left(\mathbb{E} \left[U^\alpha \left| X^{(1)} - Y^{(1)} \right| \right] + \mathbb{E} \left[(1-U)^\alpha \left| X^{(2)} - Y^{(2)} \right| \right] \right) \quad (U, 1-U \geq 0) \\ &= \frac{2u}{\alpha+1} \mathbb{E} [|X - Y|]. \end{aligned}$$

We turn to the second term. Recall that $|\max(a, b) - \max(c, d)| \leq \max(|a - c|, |b - d|)$ and $\max(a, b) = \frac{1}{2}(a + b + |a - b|)$ for any real numbers a, b, c, d . We have

$$\begin{aligned} &\mathbb{E} \left[\left| J \left(\max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) - \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)}) \right) \right| \right] \\ &\leq v \mathbb{E} \left[\max \left(\left| U^\alpha (X^{(1)} - Y^{(1)}) \right|, \left| (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right) \right] \\ &= \frac{v}{2} \left(\mathbb{E} [U^\alpha] \mathbb{E} [|X^{(1)} - Y^{(1)}|] + \mathbb{E} [(1-U)^\alpha] \mathbb{E} [|X^{(2)} - Y^{(2)}|] \right) \\ &\quad + \mathbb{E} \left[\left| U^\alpha (X^{(1)} - Y^{(1)}) - (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right] \\ &= \frac{v}{\alpha+1} \mathbb{E} [|X - Y|] + \frac{v}{2} \mathbb{E} \left[\left| U^\alpha (X^{(1)} - Y^{(1)}) - (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right] \end{aligned}$$

Combining the bounds for both terms, we get

$$\ell_1(T(\mu), T(\nu)) \leq \frac{2u}{\alpha+1} \mathbb{E} [|X - Y|] + J \mathbb{E} [|X - Y|].$$

Since $\frac{2u}{\alpha+1} + J < 1$, the operator T is a contraction in the $(\mathcal{M}_1, \ell_1)$ space.

□

Asymptotic Analysis of Some Sequences

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In this section, we analyze the asymptotic behavior of lower bounds of $\mathbb{E}[Y_n]$, as follows.

$$x_{n+2} = \frac{a}{n} \sum_{i=1}^n x_i + b, \quad n \geq 1, \quad x_1 = x_2 = 0. \quad (2.1)$$

$$x_{n+2} = \frac{a}{n} \sum_{i=1}^n x_i + \frac{b}{n} \sum_{i=1}^n |x_i - x_{n+1-i}| + c, \quad n \geq 1, \quad x_1 = x_2 = 0 \quad (a > 1, 0 < b < 1, c > 0) \quad (2.2)$$

$$x_n = \frac{2u}{n} \sum_{i=0}^{n-1} x_i + \frac{2v}{n} \sum_{i=\lfloor (n-1)/2 \rfloor}^{n-1} x_i, \quad x_0 = x_1 = 1 \quad (u, v > 0, u + v \leq 1, 2u + v > 1). \quad (2.3)$$

Proposition 3. *Consider recurrence (2.1). We have*

- If $a < 1$, then $x_n \sim \frac{b}{1-a}$.
- If $a = 1$, then $x_n \sim b \ln n$.
- If $a > 1$, then $x_n \sim \frac{b}{(a-1)\Gamma(a)} n^{a-1}$.

Proof. Consider the generating function $X(z) = \sum_{n=1}^{\infty} x_n z^n$ with the convention that $x_0 = 0$. We rewrite the recurrence as

$$(n+2)x_{n+2} - 2x_{n+2} = a \sum_{i=1}^n x_i + bn.$$

We multiply both sides by z^{n+2} and sum over $n \geq 1$. The left-hand side becomes

$$\begin{aligned} \sum_{n=1}^{\infty} ((n+2)x_{n+2} - 2x_{n+2})z^{n+2} &= \sum_{n=3}^{\infty} nx_n z^n - 2 \sum_{n=3}^{\infty} x_n z^n \\ &= z \sum_{n=1}^{\infty} nx_n z^{n-1} - 2 \sum_{n=1}^{\infty} x_n z^n \\ &= zX'(z) - 2X(z). \end{aligned}$$

The right-hand side becomes

$$\begin{aligned} \sum_{n=1}^{\infty} (a \sum_{i=1}^n x_i + bn)z^{n+2} &= az^2 \sum_{n=1}^{\infty} \sum_{i=1}^n x_i z^n + bz^2 \sum_{n=1}^{\infty} nz^n \\ &= az^2 \frac{X(z)}{1-z} + \frac{bz^3}{(1-z)^2}. \end{aligned}$$

Therefore, the generating function $X(z)$ satisfies the following ODE

$$X'(z) - \left(\frac{2}{z} + \frac{az}{1-z} \right) X(z) = \frac{bz^3}{(1-z)^2}.$$

We have $\mu(z) = \exp \left(\int - \left(\frac{2}{z} + \frac{a}{1-z} - a \right) dz \right) = z^{-2}(1-z)^a$. which gives

$$z^{-2}(1-z)^a X(z) = \int b(1-z)^{a-2} dz.$$

There are two cases of a that determine the integral. If $a = 1$, we have

$$z^{-2}(1-z)X(z) = b \ln \left(\frac{1}{1-z} \right) + C.$$

Since $x_1 = x_2 = 0$, we have $z^{-2}X(z) \rightarrow 0$ as $z \rightarrow 0$, which gives $C = 0$. Therefore, we have

$$X(z) = b \frac{z^2}{1-z} \ln \left(\frac{1}{1-z} \right).$$

Hence, $x_n = bH_{n-2} \sim b \ln n$.

If $a \neq 1$, we have

$$z^{-2}(1-z)^a X(z) = -\frac{b}{a-1}(1-z)^{a-1} + C.$$

Using the same argument as above, we have $C = \frac{b}{a-1}$. Therefore, we have

$$X(z) = \frac{b}{a-1} z^2 \left((1-z)^{-a} - (1-z)^{-1} \right).$$

Using the generalized binomial theorem

$$(1-z)^{-a} = \sum_{n=0}^{\infty} \binom{n+a-1}{n} z^n,$$

we have

$$x_n = \frac{b}{a-1} \left(\binom{n-3+a}{n-2} - 1 \right) \sim \frac{b}{a-1} \left(\frac{n^{a-1}}{\Gamma(a)} - 1 \right).$$

If $a < 1$ then $x_n \sim \frac{b}{1-a}$. If $a > 1$, then $x_n \sim \frac{b}{(a-1)\Gamma(a)} n^{a-1}$. □

Lemma 1. *The sequence (x_n) in (2.2) is increasing.*

Proof. One checks that $x_n \geq 0$ for all n . Let us prove that (x_n) is increasing. We have $x_2 \geq x_1$.

Assume that $x_n \geq x_{n-1} \geq \dots \geq x_0$ for some $n \geq 3$. Then,

$$\begin{aligned}
nx_{n+2} &= a \sum_{i=1}^n x_i + b \sum_{i=1}^n |x_i - x_{n+1-i}| + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= (a+b) \sum_{i=1}^n x_i - 2b \sum_{i=1}^{\lfloor n/2 \rfloor} x_i - bm_n + nc,
\end{aligned}$$

where

$$m_n = \begin{cases} x_{\lfloor n/2 \rfloor + 1} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even.} \end{cases}$$

Let $S_n = \sum_{i=1}^n x_i$. Then,

$$\begin{aligned}
nx_{n+2} &= (a+b)S_n - 2bS_{\lfloor n/2 \rfloor} - bm_n + nc, \\
S_{n+1} &= S_n + x_{n+1}.
\end{aligned}$$

Hence,

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2b(S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor}) - b(m_n - m_{n-1}) + c.$$

Let us check the right-hand side according to the parity of n .

- If $n = 2k$, then $S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor} = x_k$ and $m_n - m_{n-1} = -x_k$. The right-hand side is $(a+b)x_n - 2bx_k + c$.
- If $n = 2k+1$, then $S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor} = 0$ and $m_n - m_{n-1} = x_{k+1}$. The right-hand side is $(a+b)x_n - 2bx_{k+1} + c$.

Collecting the two cases, we have

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2bx_{\lceil n/2 \rceil} + c.$$

The proof follows from Proposition ??.

□

Lemma 2. *The sequence in (2.2) can be rewritten as*

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2bx_{\lceil n/2 \rceil} + c. \quad (2.4)$$

Proposition 4. *Consider the sequence defined by (2.3). We have $x_n = \Theta(n^\alpha)$, where α is the unique solution of*

$$2(u+v) - v2^{-\alpha} = \alpha + 1.$$

Proof. For the lower bound, assume the inductive hypothesis $x_n \geq K(n^\alpha + 1)$ with $K \leq \frac{1}{2}$ to be determined. The base cases $n = 0, 1$ hold. Let us check for $n = 2$.

$$x_2 = 2u + 2v = \alpha + 1 + v2^{-\alpha} \geq \alpha + 1 \geq \frac{1}{2}(2^\alpha + 1), \forall \alpha \in [0, 1].$$

For $n \geq 3$, we have

$$\begin{aligned} \frac{x_n}{K} &\geq \frac{2u}{n} \sum_{i=0}^{n-1} (i^\alpha + 1) + \frac{2v}{n} \sum_{i=\lfloor (n-1)/2 \rfloor}^{n-1} (i^\alpha + 1) \\ &\geq \frac{2u}{n} \left(\left(\int_0^{n-1} x^\alpha dx \right) + n \right) + \frac{2v}{n} \int_{(n-3)/2}^{n-1} (x^\alpha + 1) dx \\ &\geq \frac{2u}{n} \left(\left(\int_0^{n-1} x^\alpha dx \right) + n \right) + \frac{2v}{n} \left(\int_{(n-2)/2}^{n-1} x^\alpha + \frac{n+1}{2} \right) dx \\ &= \frac{2u}{n} \left(\frac{(n-1)^{\alpha+1}}{\alpha+1} + n \right) + \frac{2v}{n} \left(\frac{(n-1)^{\alpha+1} - (n/2)^{\alpha+1}}{\alpha+1} + \frac{n+1}{2} \right) \\ &= \frac{(n-1)^{\alpha+1}}{n(\alpha+1)} (2u + 2v - v2^{-\alpha}) + 2u + v \left(1 + \frac{1}{n} \right) \\ &\geq n^\alpha \left(1 - \frac{1}{n} \right)^{\alpha+1} + 2u + v \end{aligned}$$

We are done if we can prove that for every $n \geq 3$,

$$n^\alpha \left(1 - \frac{1}{n} \right)^{\alpha+1} + 2u + v \geq n^\alpha + 1.$$

By Bernoulli's inequality, we have $\left(1 - \frac{1}{n} \right)^{\alpha+1} \geq 1 - \frac{\alpha+1}{n}$. Hence, it suffices to show that

$$n^\alpha \left(1 - \frac{\alpha+1}{n} \right) + 2u + v \geq n^\alpha + 1 \iff 2u + v - 1 \geq \frac{\alpha+1}{n^{1-\alpha}}.$$

Let $N_0 = \left\lceil \left(\frac{2u+v-1}{\alpha+1} \right)^{\frac{1}{\alpha-1}} \right\rceil$. The inequality holds for every $n \geq N_0$. Now we choose K for it to hold for $n < N_0$. In particular,

$$K = \min \left(\frac{1}{2}, \min_{3 \leq i < N_0} \frac{x_i}{i^\alpha} \right)$$

The lower bound follows. □

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